

PART II: BASIC ELECTRONICS ENGINEERING

Max. Marks: 50

Duration: 90 min

PART A

Answer all questions, each carries 4 marks.

- 12 What are the different types of capacitors? Find the value of the capacitor coded as 103.
- 13 Sketch the energy band diagram of conductors, insulator and semiconductors.
- 14 With a neat diagram explain the working of an instrumentation system.
- 15 What is biasing? List the advantages of potential divider biasing.
- 16 Discuss the need for modulation.

(5×4=20)

PART B

Answer one full question from each module, each question carries 10 marks

Module-IV

- 17 a) Give the specifications of a resistor. The colour bands marked on a resistor are Green, Blue, Orange and Gold. What are the minimum and maximum resistance values expected from that resistance? (5)
- b) Explain the working of a diode under forward and reverse biased condition. Draw its VI characteristics? (5)

OR

- 18 a) Differentiate between avalanche breakdown and zener breakdown. (4)
- b) Narrate the working of an NPN transistor. (6)

Module-V

- 19 a) Explain the working of a bridge rectifier. (5)
- b) Describe the block diagram of a public addressing system. (5)

OR

- 20 a) Define line regulation and load regulation. (4)
- b) Draw the frequency response curve of a CE amplifier and explain. (6)

Module-VI

- 21 a) Describe the basic principles of a cellular communication system. (5)
- b) Write the expression for an AM wave. Draw the frequency spectrum and find the associated bandwidth? (5)

OR

- 22 a) What are the advantages of GSM network? (4)
- b) Explain the roles of BTS, BSC and MSC in a GSM network. (6)

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S1 (Special Improvement) Examination January 2021 (2019 scheme)

Course Code: EST130

Course Name: **BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING**
(2019-Scheme)**PART I: BASIC ELECTRICAL ENGINEERING**

Max. Marks:50

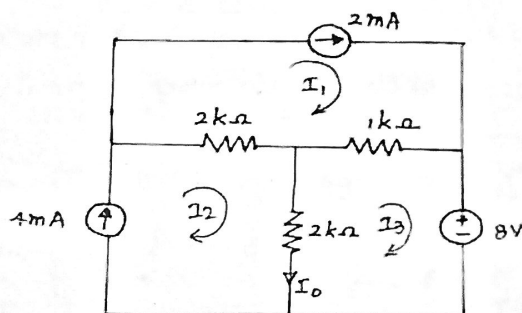
Duration: 90 min

PART A*Answer all questions, each carries 4 marks.*

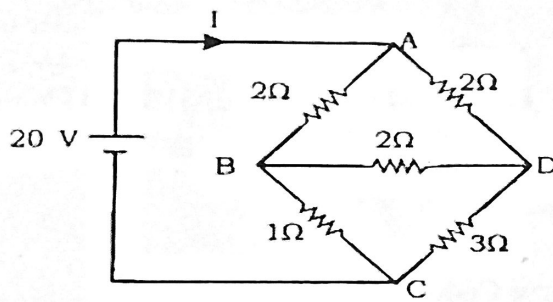
- 1 Derive the expression for rms value of a sinusoidal wave form.
- 2 Define self inductance of a coil and derive an expression for the same.
- 3 Derive an expression for energy stored in a capacitor.
- 4 Prove that in a purely inductive circuit, current lags behind the applied voltage by 90 degrees and the power consumed is zero.
- 5 With the help of circuit diagram and phasor diagram, derive the relation $(5 \times 4 = 20)$ between line and phase voltage in a three phase star connected system.

PART B*Answer one full question from each module, each question carries 10 marks***Module-I**

- 6 Find I_o in the circuit using mesh current analysis. (10)

**OR**

- 7 Find the source current I in the figure using star-delta transformation. (10)

**Module-II**

- 8 a) An iron ring of mean diameter 10cm is uniformly wound with a coil of 2000 turns. When a current of 0.25A is passed through the coil a flux density of 0.4T is set up in the iron ring. Find i) the magnetizing force and ii) the relative permeability of iron. (4)
- b) Define i) self-inductance of a coil ii) coefficient of coupling and iii) relative permeability (6)

OR

- 9 Two coils A and B have 12000 turns and 14000 turns respectively. 80% of the flux produced by coil A links with coil B. A current of 6A in coil A produces 0.05mwb in coil A while the same current in coil B produces 0.085mwb in coil B. Calculate i) Mutual inductance and ii) Coefficient of coupling. (10)

Module-III

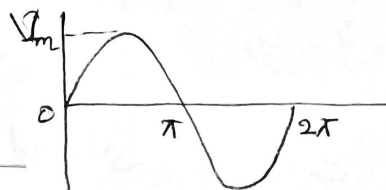
- 10 Coil A having resistance of $20\ \Omega$ and inductance of 0.2 H is connected in series with another coil B having resistance of $15\ \Omega$ and inductance of 0.1H. The two coils in series are fed from 220V, 50 Hz, single phase power supply. Determine (i) the voltage across each coil (ii) power dissipated in each coil (iii) power factor of the whole circuit. (10)

OR

- 11 Three similar coils connected in star draw a total power of 1.5kW at a power factor of 0.2 lagging from a 3 phase 400V, 50Hz power supply. Calculate the resistance and inductance of each coil (10)

1. RMS Value of Sinusoidal wave form.

V_{rms} = $\frac{\text{Area of the first half cycle of the squared wave}}{\text{It's base}}$



$$V_{rms}^2 = \frac{\int_0^\pi V_m^2 \sin^2 \theta \cdot d\theta}{\pi} = \frac{1}{\pi} \int_0^\pi V_m^2 \sin^2 \theta \cdot d\theta = \frac{V_m^2}{\pi} \int_0^\pi \frac{1 - \cos 2\theta}{2} \cdot d\theta$$

$$= \frac{V_m^2}{\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi = \frac{V_m^2}{2}$$

$$\therefore V_{rms} = \sqrt{\frac{V_m^2}{2}} = \frac{V_m}{\sqrt{2}} = \underline{\underline{0.707 V_m}}$$

2. Self inductance of a coil (L)

It is the property of a coil which consists in opposing any change, increase or decrease, of current and flux through it, is called its self inductance. This property of a coil is analogous to inertia in a mechanical body. So it is also termed as electrical inertia or electromagnetic inertia.

Expression for self inductance

Let N - No. of turns of the coil

I - Current through the coil

ϕ - the resulting flux linkage with the coil.

Total flux linkage $N\phi$ is proportional to I .

So induced emf, $e = -N \frac{d\phi}{dt} = -\frac{d(N\phi)}{dt}$

$$e \propto -\frac{dI}{dt}$$

$$\therefore e = -L \frac{dI}{dt} \quad \therefore \underline{\underline{L = -\frac{e}{dI/dt}}}$$

$$\therefore \frac{N d\phi}{dt} = \frac{L dI}{dt} ; N\phi = LI$$

$$\therefore L = \frac{N\phi}{I}$$

3. Energy stored in the capacitor.

Consider the process of charging of a capacitor to a voltage of V volts. Let the voltage be v at some stage of charging. This equals the work done in moving one Coulomb of charge from one plate to the other. Let dq be the charge transferred. Then this additional work done is

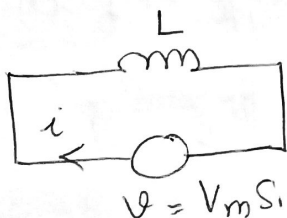
$$dW = v \cdot dq, \text{ but } q = Cv, \quad dq = C \cdot dv$$

$$\therefore dW = C \cdot v \cdot dv$$

So total work done in creating V units of potential is

$$W = \int_0^V dW = \int_0^V C \cdot v \cdot dv = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C}$$

4.



Let $v = V_m \sin \omega t$ be the voltage applied across a pure inductor of inductance L henry.

Let the current flowing through the coil be i .

We know

$$v = L \frac{di}{dt}$$

$$v = V_m \sin \omega t \quad \text{--- (1)}$$

$$\therefore \frac{di}{dt} = \frac{v}{L}$$

$$\int \frac{di}{dt} \cdot dt = \int \frac{v}{L} \cdot dt$$

$$i = \int \frac{V_m \sin \omega t}{L} \cdot dt = \frac{V_m}{\omega L} [-\cos \omega t]$$

$$i = \int \frac{V_m \sin \omega t}{L} \cdot dt = \frac{V_m}{\omega L} [-\cos \omega t]$$

(2)

$$i = \frac{V_m}{L\omega} \sin(\omega t - 90^\circ) = I_m \sin(\omega t - 90^\circ) \quad (2)$$

Comparing (1) and (2),
current lags the voltage by 90° .

Power

$$\begin{aligned} \text{Instantaneous power } p &= v i \\ &= V_m \sin \omega t \cdot I_m \sin(\omega t - 90^\circ) \\ &= V_m I_m \sin \omega t \times -\cos \omega t \\ &= -\frac{V_m I_m}{2} \sin 2\omega t \end{aligned}$$

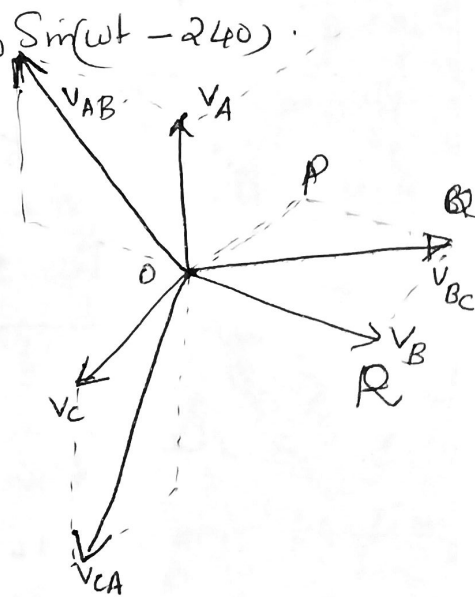
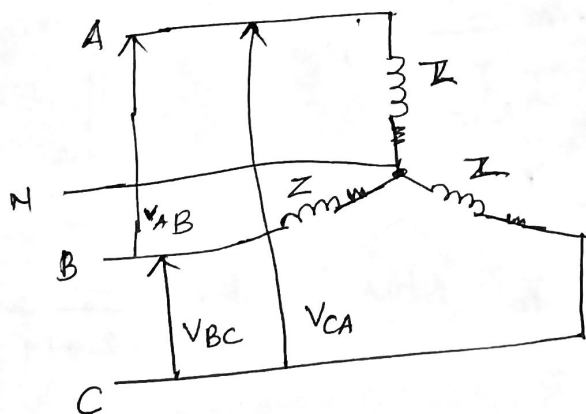
Over a complete cycle $\sin 2\omega t = 0$.

$$\therefore \underline{\underline{p = 0}}$$

5 Consider a three phase circuit with voltages

$$V_A = V_m \sin \omega t$$

$$V_B = V_m \sin(\omega t - 120^\circ) \text{ and } V_C = V_m \sin(\omega t - 240^\circ)$$



Let $\bar{V}_A$, $\bar{V}_B$ and $\bar{V}_C$ represents the phase values, and V_{AB} , V_{BC} and V_{CA} be the line values of voltages.

$$\text{we know, } V_{AB} = V_{AN} - V_{BN} = V_A - V_B \text{ and}$$

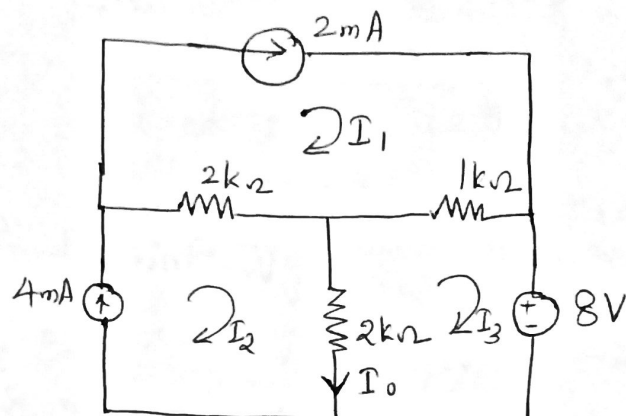
$$V_{BC} = V_B - V_C \text{ and } V_{CA} = V_C - V_A$$

Considering the parallelogram OPQR,

$$OQ = V_L = V_{BC} = \sqrt{V_A^2 + V_B^2 + 2V_A V_B \cos 60^\circ} = \underline{\underline{\sqrt{3} V_P}}$$

Where V_p - the phase value of voltage
 V_L - the line value of voltage.

6.



$I_1 = 2 \text{ mA}$ (Since
 $I_2 = 4 \text{ mA}$ current sources)

$$1k(I_3 - I_1) + 8 + 2k(I_3 - I_2) = 0$$

$$3kI_3 - 2kI_2 - 1kI_1 + 8 = 0$$

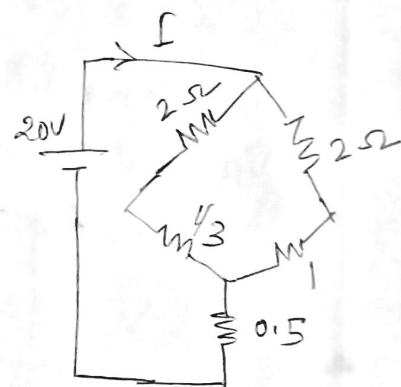
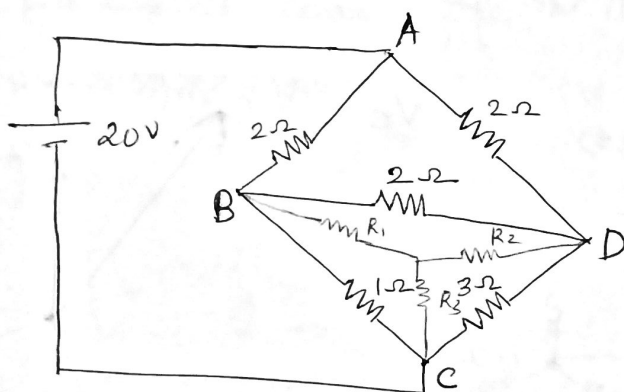
$$I_0 = I_2 - I_3$$

~~$3 \times 10^3 \times 4 \times 10^{-3} - 2 \times 10^3 \times 2 \times 10^{-3} - 1 \times 10^3 \times 2 \times 10^{-3} + 8 = 0$~~

$$4, \quad 3kI_3 - 2 \times 10^3 \times 4 \times 10^{-3} - 2 \times 10^3 \times 1 \times 10^{-3} + 8 = 0$$

$$3kI_3 - 8 - 2 + 8 = 0 \quad ; \quad I_3 = \frac{2}{3} \text{ mA} \quad ; \quad I_0 = 4 - \frac{2}{3} = \underline{\underline{\frac{10}{3} \text{ mA}}}$$

7.



Convert ΔBCD to star, $R_1 = \frac{2 \times 1}{2 + 1 + 3} = \frac{2}{6} = \underline{\underline{\frac{1}{3} \Omega}}$

$$R_2 = \frac{3 \times 2}{6} = 1 \Omega, \quad R_3 = \frac{1 \times 3}{6} = 0.5 \Omega$$

$$\therefore R_{eq} = \left(2 + \frac{1}{3}\right) \parallel \frac{1}{3} + 0.5$$

$$= \underline{\underline{1.8125 \Omega}}$$

$$\therefore \text{Current supplied} = \frac{20}{1.8125} = \underline{\underline{11.03 \text{ A}}}$$

$$8. a) d = 10 \times 10^{-2} \text{ m}$$

$$N = 2000$$

$$I = 0.25 \text{ A}$$

$$B = 0.4 \text{ T} ; H = ? \quad \mu_r = ?$$

$$H = \frac{NI}{l} = \frac{2000 \times 0.25}{\pi \times 10 \times 10^{-2}} = \underline{\underline{1592 \text{ A/m}}}$$

$$B = \mu_0 \mu_r H$$

$$\therefore \mu_r = \frac{B}{\mu_0 H} = \frac{0.4}{4\pi \times 10^{-7} \times 1592} = \underline{\underline{200}}$$

b) Self inductance of a coil. ref. answer ①.

cc) Coefficient of coupling (K)

Coefficient of coupling may be defined as the ratio of mutual inductance actually present between the coils to the maximum possible value. If the entire flux due to one coil links with the other coil, then the value of coefficient of coupling k is unity. If the flux of one coil does not link at all with the other coil, then $k=0$.

$$k = \frac{M}{\sqrt{L_1 L_2}} \quad \begin{array}{l} M - \text{mutual inductance b/w } L_1 \text{ \& } L_2 \\ L_1, L_2 - \text{self inductances of coil 1 \& 2} \end{array}$$

ccc) Relative permeability (μ_r)

μ_r of a medium or material is defined as the ratio of the flux density produced in that medium or material to the flux density produced in vacuum by the same magnetising force.

$$\mu_r = \frac{\text{Flux density in the medium}}{\text{Flux density in Vacuum}}$$

$$9. N_A = 12000$$

$$N_B = 14000$$

$$\phi_B = 0.8 \phi_A$$

$$I_A = 6A$$

$$\phi_A = 0.05 \times 10^{-3}$$

$$\phi_B = 0.085 \times 10^{-3} \text{ if } I_B = 6A.$$

$$\begin{aligned} i) M &= \frac{k \phi_A N_B}{I_A} = \frac{0.8 \times 0.05 \times 10^{-3} \times 14000}{6} \\ &= \underline{0.093H} \end{aligned}$$

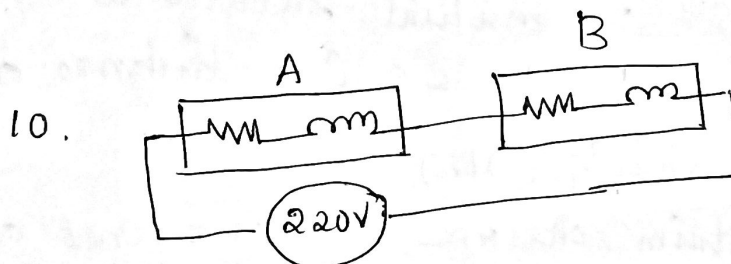
$$ii) K = ? \quad K = \frac{M}{\sqrt{L_1 L_2}}$$

$$L_1 = \frac{N_A \phi_A}{I_A} \text{ and } L_2 = \frac{N_B \phi_B}{I_B}$$

$$L_1 = \frac{12000 \times 0.05 \times 10^{-3}}{6} = 0.1H$$

$$L_2 = \frac{14000 \times 0.085 \times 10^{-3}}{6} = 0.198H$$

$$\therefore K = \frac{0.093}{\sqrt{0.1 \times 0.198}} = \underline{\underline{0.66}}$$



$$R_A = 20\Omega, L_A = 0.2H$$

$$R_B = 15\Omega, L_B = 0.1H$$

(4)

$$Z_A = R_A + jX_{LA}$$

$$X_{LA} = 2\pi f L_A = 2\pi \times 50 \times 0.2 = 62.83 \Omega$$

$$\therefore Z_A = 20 + j 62.83 = \underline{65.93 \angle 72.34^\circ \Omega}$$

$$Z_B = R_B + jX_{LB}$$

$$X_{LB} = 2\pi f L_B = 2\pi \times 50 \times 0.1 = 31.41 \Omega$$

$$Z_B = 15 + j 31.41 \Omega = \underline{34.81 \angle 64.47^\circ \Omega}$$

$$\begin{aligned} \therefore \text{Total impedance of the circuit} &= Z_A + Z_B \\ &= (20 + j 62.83) + (15 + j 31.41) \\ &= 35 + j 94.25 \Omega = \underline{100.53 \angle 69.62^\circ \Omega} \end{aligned}$$

$$\text{Total current} = \frac{V_{\text{applied}}}{\text{Total Impedance}}$$

$$= \frac{220}{100.53 \angle 69.62} = \underline{\underline{2.18 \angle -69.62^\circ \text{ A}}}$$

$$\begin{aligned} \text{Voltage across coil A} &= I Z_A \\ &= 2.18 \times 65.93 \\ &= \underline{\underline{143.72 \text{ V}}} \end{aligned}$$

$$\begin{aligned} \text{Voltage across coil B} &= I Z_B \\ &= 2.18 \times 34.81 = \underline{\underline{75.88 \angle -5.45^\circ \text{ V}}} \end{aligned}$$

$$\begin{aligned} \text{Power absorbed in coil A} &= I^2 R_A = 2.18^2 \times 20 \\ &= \underline{\underline{95.048 \text{ W}}} \end{aligned}$$

$$\begin{aligned} \text{Power absorbed in coil B} &= 2.18^2 \times 15 \\ &= \underline{\underline{71.286 \text{ W}}} \end{aligned}$$

11. Total power = 1.5 kW at a pf of 0.2 lagging.

$$4, \sqrt{3} V_L I_L \cos \phi = 1.5 \times 10^3$$

in star $I_L = I_p$

$$V_L = 400 \text{ V}$$

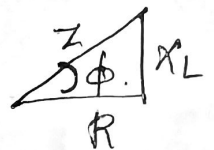
$$\cos \phi = 0.2$$

$$4, \sqrt{3} \times 400 \times I_L \times 0.2 = 1.5 \times 10^3$$

$$\therefore I_L = \frac{1.5 \times 10^3}{\sqrt{3} \times 400 \times 0.2} = 10.838 \angle -78.46^\circ \text{ A}$$

$$Z_p = \frac{V_p}{I_p} = \frac{400}{10.838} = 21.38 \angle 78.46^\circ \Omega$$

$$\cos \phi = R/Z$$



$$\therefore R = Z \cos \phi = 21.38 \times 0.2 = 4.268 \Omega$$

$$X = Z \sin \phi = 21.38 \times \sin(\cos^{-1} 0.2) = 20.908 \Omega$$

$$X_L = 2\pi f L \quad \therefore L = \frac{X_L}{2\pi f} = \frac{20.908}{2\pi \times 50} = 0.066 \text{ H}$$